4. For each of the following say whether it is an example of a chemical change or a physical change. Give a reason for your answer. If you are not sure, explain why you are having difficulties making a decision.

Example of change	Physical or chemical	Reasons
Dissolving sugar in a cup of hot coffee		
Burning fallen leaves		
Melting wax		
Making yoghurt from milk		
Paint drying		
Breaking an egg		
Boiling an egg		
Letting off fireworks		

Chemical or physical change?

1. What are the characteristics of a physical change?

2. What are the characteristics of a chemical change?

3. What evidence would you look for that would help you decide that a change was a chemical change?

Think

Vinegar is the common name for acetic acid. When bicarbonate of soda is added to an acid, a gas is produced.

9. Bicarbonate of soda is one of the ingredients in baking powder. The other ingredient is tartaric acid. Baking powder is added to cakes to make them rise. Use your knowledge of chemical changes to explain why cakes rise.

10. In a container of baking powder is a mixture of bicarbonate of soda and tartaric acid. There is no reaction in the container but there is a reaction when baking powder is added to the ingredients for a cake. Why do you think this happens?

Extension

11. If you can get hold of some baking powder, test your idea.

What happened when you tested your idea?



Record of mass

5. Record the mass of the equipment in this table.

Mass of the equipment (g)		
Start of activity	End of activity	Change in mass

6. If the mass had increased, this would mean that matter had been added to the equipment. How could this have happened?

7. If the mass had decreased, this would mean that matter had been lost from the equipment. How could this have happened?

8. What does it mean if there was no change in mass?

Chemical change

1. What was the aim of this activity?

_____

_____

Observations

2. Record the changes you observed.

_____

_____

_____

_____

3. Were these changes evidence of a physical change or a chemical change? Give reasons for your response.

_____

_____

_____

_____

_____

_____

Complete these sentences

4. When _____ is added to vinegar a _____ change takes place. The evidence of a _____change is a _____ being produced.

CHEMICAL SCIENCE @ work: 3.2

What to do:

1. Place a teaspoonful of bicarbonate of soda in the centre of the paper tissue. Gather up the edges of the tissue and tie it with thread to make a 'bag' containing the powder. Make sure there is no bicarbonate of soda on the outside of the bag.
2. Fill the plastic bottle a third full with vinegar.
3. Suspend the bag inside the bottle above the vinegar. Drape the thread so that it hangs outside the bottle and hold it in place with the elastic band. Make sure that the bag and the vinegar do not touch.
4. Blow up the balloon a little way and let the air out. Do this a couple of times to stretch the balloon. Stretch the open end of the balloon over the opening of the bottle.
5. If you have access to a balance, record the mass of the equipment at this stage.
6. Gently shake the bottle so that the vinegar soaks the paper tissue. The tissue should break down and release the bicarbonate of soda into the vinegar.
7. Observe what happens.
8. When you have completed your observations, record the mass of the equipment again.
9. Use *Worksheet 3.3* to record your observations in words and pictures. You could use Figure 3.12 as stimulus for your pictures.





Use *Worksheet 3.4* 'Chemical or physical change?' to check your understanding of physical and chemical changes.

questions 3.1

1. What is 'matter'?
2. Name three states of matter.
3. What kind of change is a change of state—physical or chemical?
4. What are two indicators of a chemical change?
5. Name two types of chemical change.

35

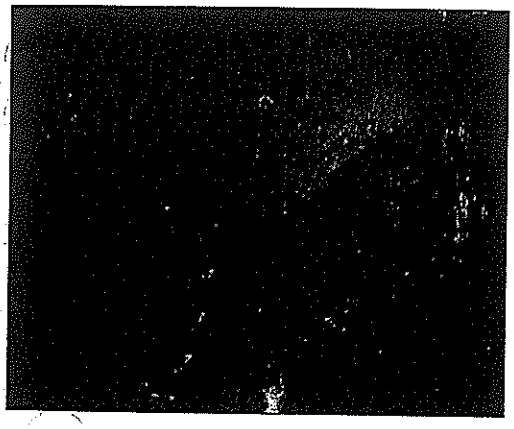


Figure 3.11
Rust is produced by a chemical change when iron and oxygen react together.

'Don't leave your bike in the rain. It will rust.' Have you heard something like that from your parents? Rusting is a chemical change. Iron and oxygen (from the air) react together to form the red, crumbly substance we call rust.

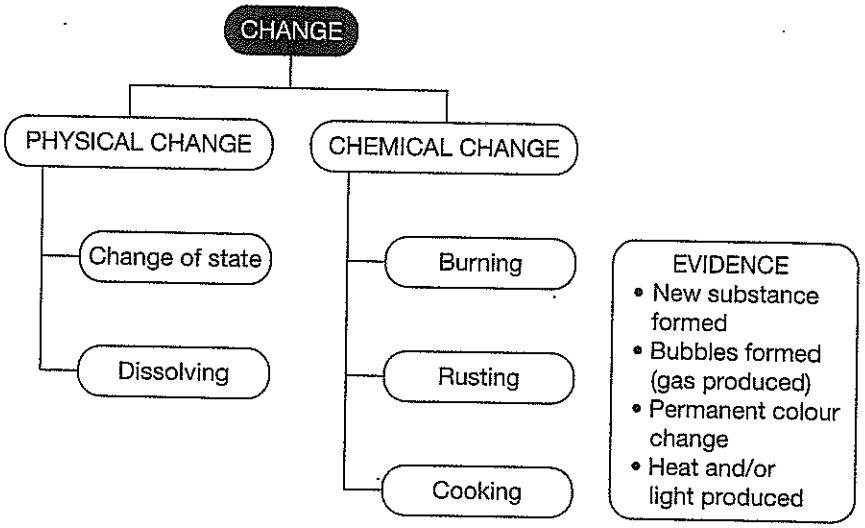


Figure 3.12
Physical and chemical changes.

CHEMICAL SCIENCE @ work 3.2

Aim: To investigate a chemical change

Materials:

wide-necked plastic bottle
vinegar
paper tissue
thread or thin string
bicarbonate of soda (baking soda)
balloon
teaspoon
elastic band
balance or scales (measuring to 1 gram)
Worksheet 3.3 'Chemical change'
safety glasses

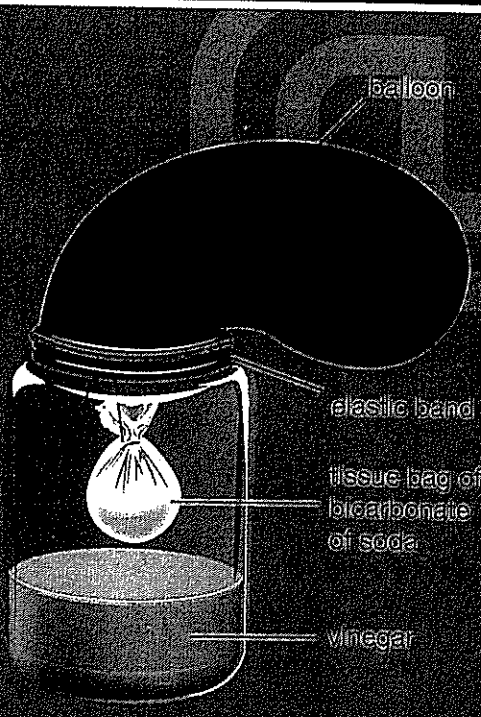


Figure 3.13
Experimental set-up

CHAPTER 12

12.1 WHAT IS A CHEMICAL REACTION?

Changes occur around us all the time. Bread goes mouldy, nails rust, milk becomes sour, seeds sprout, water evaporates . . .

These changes can be classified as either physical or chemical. During a **physical change** some of the properties of the substance change, but the substance is still the same. Also, you can usually reverse a physical change. For example, when liquid water freezes to ice its properties change. It is now a solid which cannot be poured—but it is still water. And it can easily be changed back to liquid water.

The tungsten filament in a light bulb becomes white hot when electricity passes through it, but the filament is still tungsten. This is a physical change, because *no new substance has been formed.* When you turn the light off, the filament is the same as it was before.

In a chemical change, one or more new substances are formed. Chemical changes are usually called **chemical reactions**.

Burning is a chemical reaction. In a bushfire, bushes and trees are burnt, producing smoke and invisible gases, and leaving ash (carbon). The smoke, gases and ash are new substances—completely different from those in the original bushes and trees. The reaction also produces heat and light.

The rusting of iron is another chemical reaction. It produces a new brown substance, with properties different from the original grey iron.

Chemical reactions cannot easily be reversed. If you burn something you cannot get it back; and you cannot change rust back to iron without another chemical reaction and an input of energy.

Chemistry is the study of materials and how they change. In Experiment 1 you can start a study of chemistry by observing some chemical reactions.

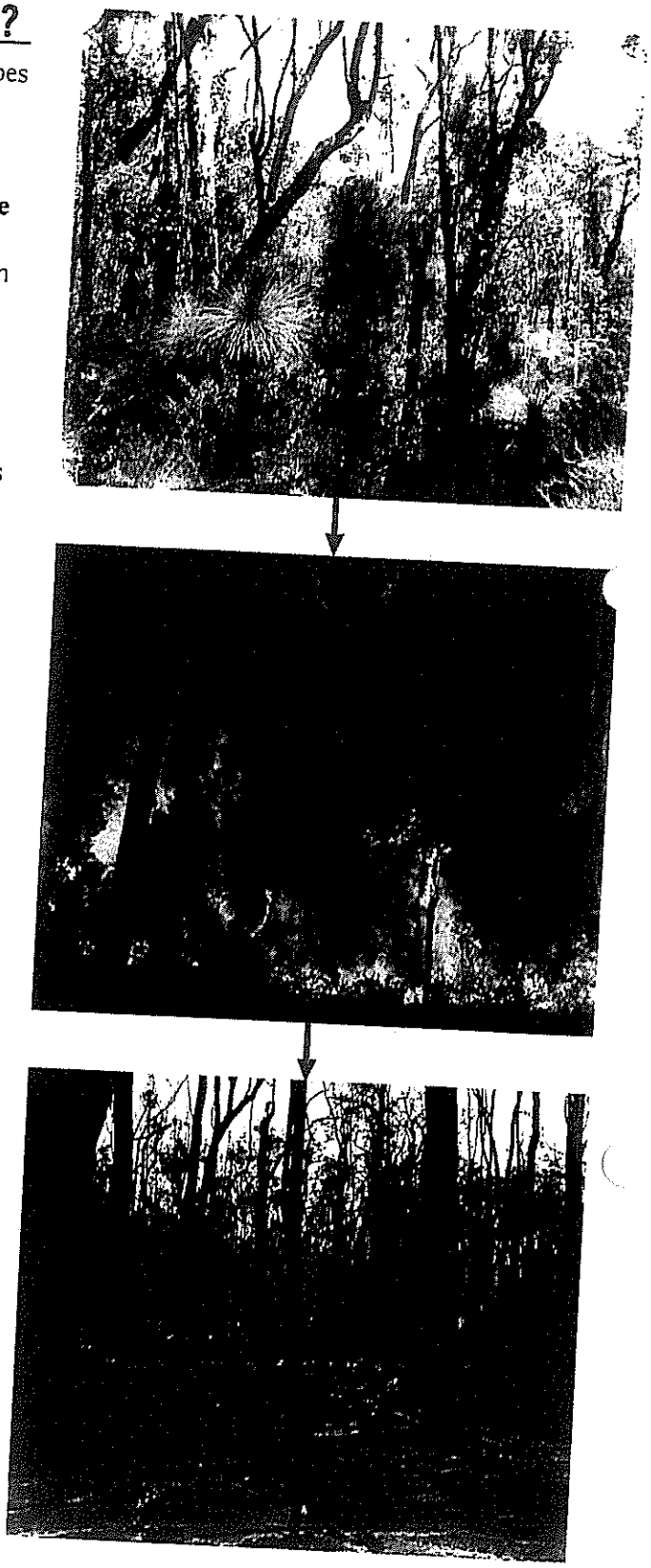


Fig 2 *Burning is a chemical reaction called combustion which produces new substances.*

ACTIVITY 2.6

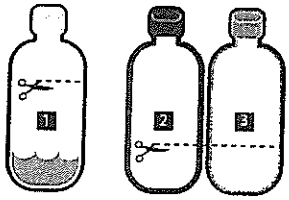
Water cycle in a bottle

Understand how the water cycle works? Now it's time to make your own!

What you need:

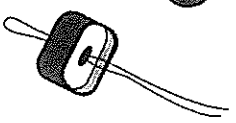
- 3 one-litre plastic drink bottles with labels removed
- potting mix or light soil
- cup of water
- fast-growing seeds such as cress or radish
- marker
- empty film cartridge
- 2 pieces of string (not nylon) —40 cm and 20 cm
- ice cubes

1



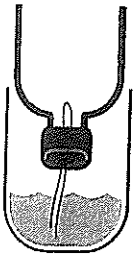
Write on the bottles 'chamber 1', 'chamber 2' and 'chamber 3'. Cut the top off chamber 1 at the level shown and pour in a cup of water. Cut the bottoms off chambers 2 and 3.

2



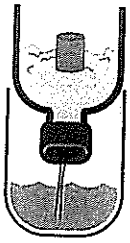
Make a hole in the cap of chamber 2, just big enough for the string to fit through. Fold the 40 cm piece of string in half and loop it through the hole. Wet the string.

3



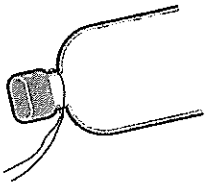
Place chamber 2 on top of chamber 1 to check that the water level is high enough for the strings to dangle into it.

4



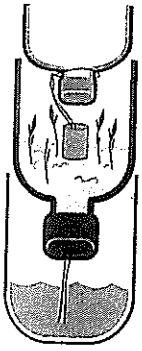
Put a cup of moist soil into chamber 2 and push the film cartridge in the middle of the soil (open end upwards). Plant 8–10 seeds in a circle around the cartridge.

5



Tie the 20 cm piece of string around the cap of chamber 3, making sure it is long enough to dangle into the film cartridge when placed on top of chamber 2. Do not assemble the chambers until you're ready to create the water cycle.

6



Wait until the seedlings are 4–5 cm tall. Wet the string thoroughly on chamber 3 and place it on top of chamber 2. Put ice cubes in chamber 3 and observe what happens.

What's going on here?

Water is drawn up from chamber 1 into the soil in chamber 2, where it is taken up by the plants through their roots. Water is released into the second chamber by evaporation from the soil and transpiration from the plants. When the water vapour hits the cold surface of chamber 3, it condenses and falls as precipitation (rain). The rainwater percolates back through the soil and into the water source below. This cycle will continue as long as chamber 3 is cold enough.

Tips and tricks:

- Don't use coloured string —the dye will run!
- If water leaks from the hole in the bottle cap, tie a knot in the string to cover it.
- Put your water cycle in a warm and sunny place.

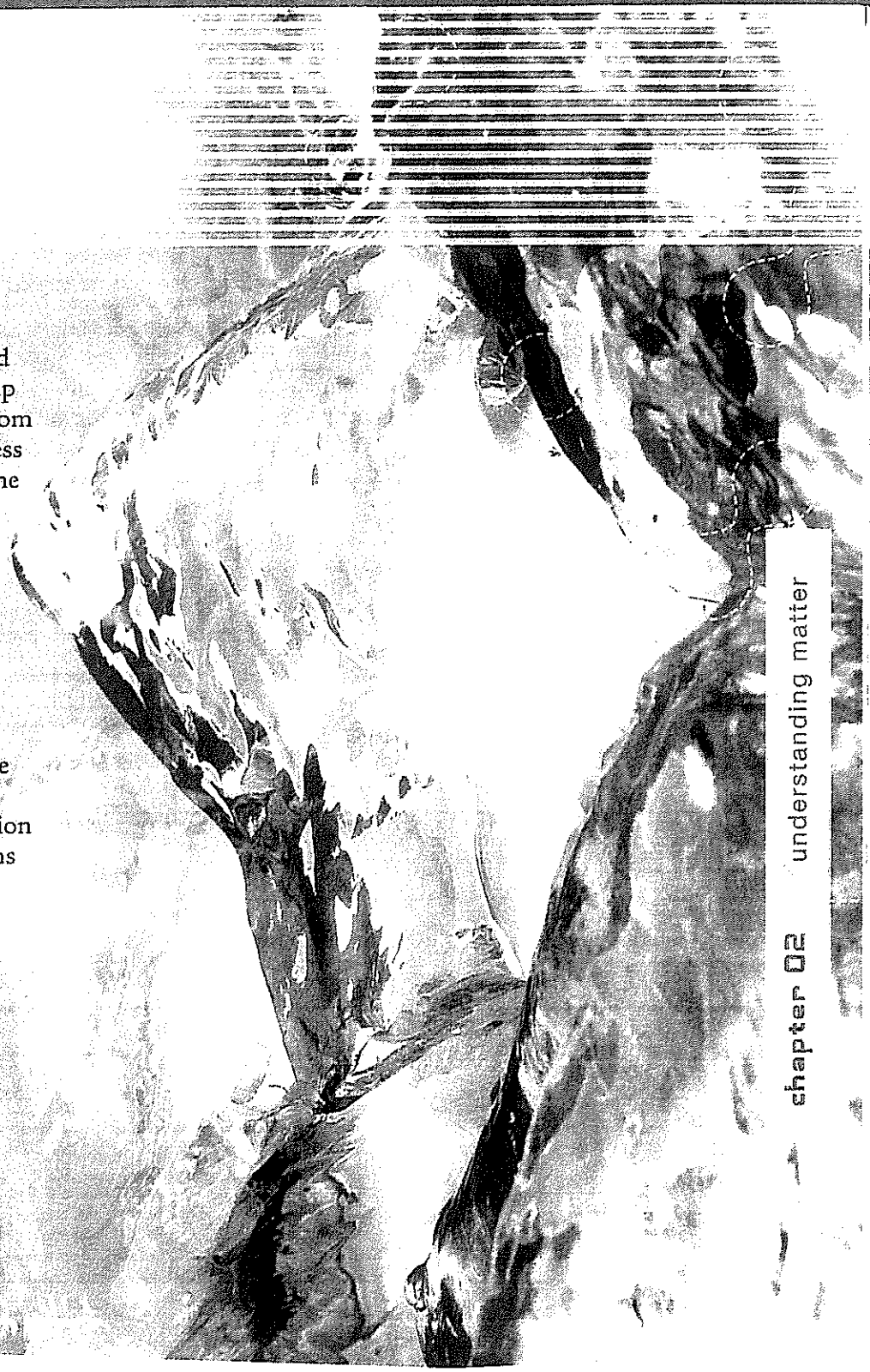
1. In the middle of a sheet of paper, draw a water cycle in a bottle. Down one side, label where each stage of the water cycle occurs in the bottle. Down the other side, label the parts of the environment where this stage occurs. Use the terms: evaporation, transpiration, condensation, precipitation and percolation.
2. Create a PMI chart of what you have learnt about the water cycle from this activity.

Inspiration

Water also cycles through living things. When it rains, water soaks down into the soil in a process called percolation. Plants take this water up through their roots. It evaporates from the leaves of a plant again in a process called transpiration, and re-enters the environment.

Animals also drink water that has fallen as rain. Eventually the water passes back through the animals as it evaporates from their bodies or is excreted as waste.

It is perfectly clean as it has been evaporated and condensed, but the water you've drunk today could have been in any of these places and in times stretching back to the formation of water on this planet! If only atoms could talk!



questions

1. What percentage of the Earth's water supply is fit to drink?
2. What is water vapour and where does it come from?
3. List and describe three ways that rain can eventually find its way back to being water vapour.
4. Would the water cycle exist on Earth if the boiling point of water was 200°C? Present your answers in a cause and effect wheel.
5. Imagine you are one molecule of water vapour in a cloud over Fraser Island. Write a journal account of one day of your holidays.

Worksheet

in WATER

Can you imagine life with no water? There is probably no need, as life on Earth couldn't exist without it!

The human body is made up of about seventy per cent water. That's a lot of clear liquid. Ninety-seven per cent of the water found naturally on Earth is salt water and found in seas. Only three per cent is fresh water and three-quarters of that is frozen. This means that less than one per cent of the Earth's water is available for our use! The amount of water that we have is finite and what we do have is continuously being recycled in a process that began a long time ago.

Energy from the Sun causes water to evaporate. The greatest amount of evaporation comes from the oceans, as they are the largest water bodies. As the water vapour rises higher, it cools the air around it. At a certain point, this air will not be able to hold any more water vapour and we say that the air is saturated. When this happens, some of the vapour will condense into tiny droplets, changing state from a gas to a liquid. These water droplets rise in the atmosphere and form clouds. As these get colder, eventually the water droplets combine to form larger droplets, which fall as rain or snow (precipitation). The precipitation that falls on land then flows into streams and rivers, and finally back into the ocean.

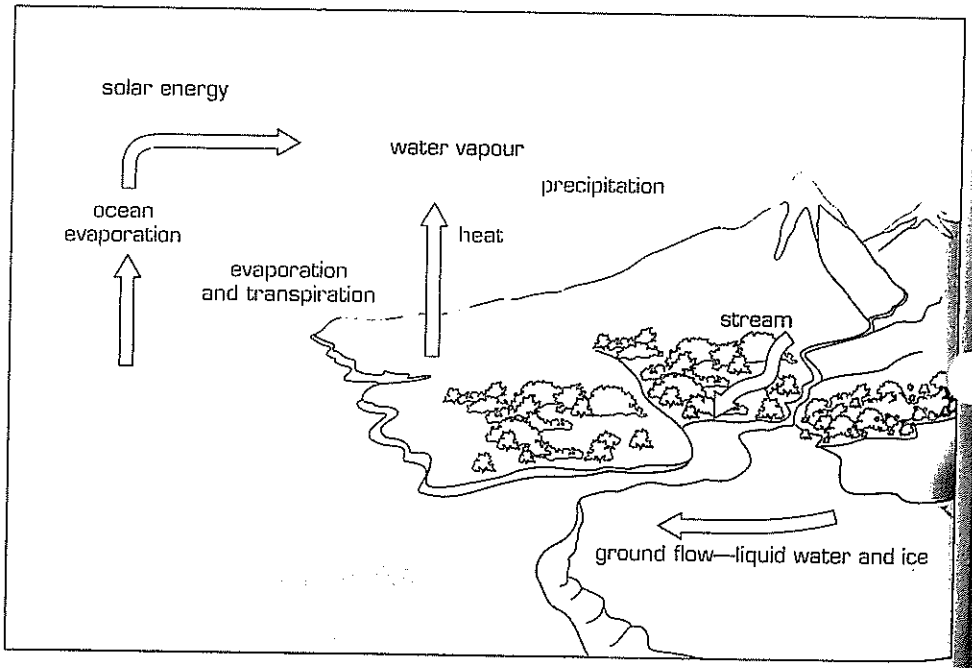


Figure 2.16
Nature recycles all of the water that exists on Earth in a system called the water cycle.

Name: _____ **Class:**_____

What do I know about changes?

Record your ideas about changes here.

Changes as I make my breakfast or lunch

What I know about the changes

These changes are reversible.

Causes of changes

These new things were produced.

Questions I have	Where I can look for information



Challenge

8. Dissolving is a physical change. This means that the change can be reversed. Outline your idea for a way of recovering a dissolved substance from the solution.

Discuss your ideas with a partner and then with the rest of the class.

9. What idea did the class come up with?

10. If you were able to test your idea, what was the result?

5. Complete the following statements.

I know that _____ dissolved in water because _____

_____.

If a substance s_____ on the b_____ or

f_____ on the t_____ when mixed with water, it is not soluble.

Write

6. Write one or two sentences that use the words 'soluble', 'insoluble' and 'solution' correctly and in a way that explains their meaning.

_____

_____

_____

_____

_____

Think

7. Suggest a way of demonstrating that the substances have dissolved and not 'disappeared'.

_____

_____

_____

_____

_____

Name: _____ **Class:** _____

Dissolving

1. What is the aim of this activity?

_____

_____

2. What has been done to make sure that this is a fair comparison?

_____

_____

3. What do you predict will happen?

_____

_____

Observations

4. Record your observations in the following table. Your observations should be one of:
   - dissolves
   - settles on the bottom
   - floats on the top.

Solid	Observation	Soluble/insoluble
Brown sugar		
Sand		
White sugar		
Ground-up chalk		
Flour		
Salt		

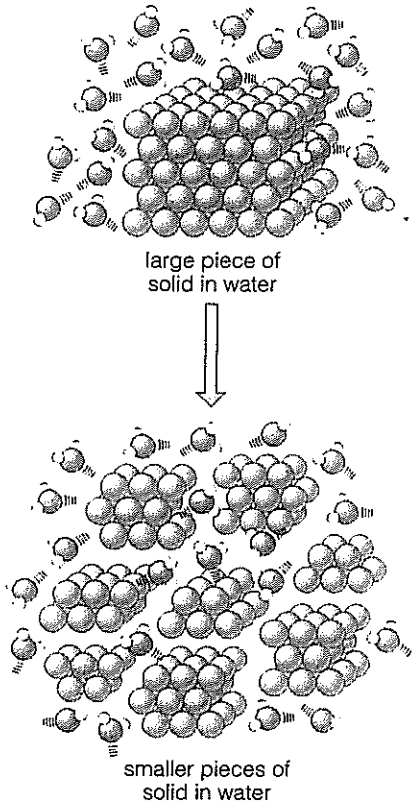


Figure 4.5 *When the solute is broken up, more of its particles are at the surface.*



Figure 4.6 *Nail varnish dissolves in propanone.*

Speeding up dissolving

There are three things that you can do to make something dissolve faster:

- heat up the solvent
- stir or shake the solvent
- grind the solute up into smaller pieces.

Heating and stirring both make the solute dissolve faster by making the solvent particles move faster, so hitting the solute particles harder and more often. In addition, heating makes the solute particles vibrate faster, helping them to break free more easily.

When a solute dissolves, the particles under the surface cannot dissolve until those above them on the surface have dissolved. When a solute is ground up into small pieces, more of the particles are at the surface, so it dissolves faster.

Different solvents

Water is the most common solvent because it is readily available and cheap, but there are many other solvents. A substance that might dissolve in one solvent, may not be soluble in another. For example, salt dissolves in water but not in ethanol (alcohol). Some substances are soluble in more than one solvent. For example, sugar dissolves in both water and ethanol.

Solvents have many uses, some of which are shown in Table 4.1.

Some people think that solvents are glues. This is wrong, but some glues work by having a sticky substance dissolved in a solvent. When the glue is used, the solvent evaporates leaving the sticky substance in place.

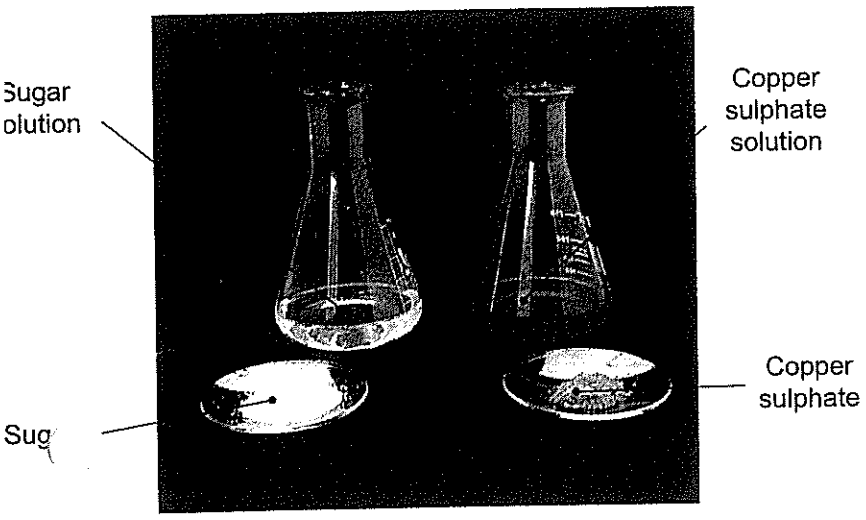
Table 4.1

Solvent	Use
propanone	nail varnish remover, removing grease
ethanol	some glues, aftershaves, perfumes
tetrachloroethene	dry cleaning clothes, cleaning circuit boards
white spirit	gloss paint, paint brush cleaner
toluene	glue remover (for some glues)

ubstance is **insoluble** if it cannot dissolve in a solvent. For example, ıd is insoluble in water. When an insoluble substance is shaken in a uid, the solid particles can be seen spread throughout the liquid. This :alled a **suspension** (you cannot see through a suspension). The solid rticles usually settle down to the bottom quite quickly.

test if a substance is soluble or insoluble in a solvent, you add a ıall amount of the substance to the solvent and then stir or shake. ter this, if the substance cannot be seen it is soluble, but if you can ll see the solid then it is insoluble.

her vhite substance dissolves, a colourless solution is formed (for ample when sugar dissolves in water). When a coloured substance ssolves in water, a coloured solution is formed (for example when pper sulphate crystals dissolve in water).



gure 4.3 *Solutions.*

s dissolving a chemical or physical change?

issolving is a **physical** change. The solute and solvent do not hange into different substances, they just mix together. For example, hen sugar dissolves in water, the solution is a mixture of sugar and ater, the same two substances. If we evaporate the water, the sugar ill be left behind.

ome substances react with water producing different substances. This s a **chemical** change and is not the same as dissolving. For example, hen sodium is added to water, a reaction takes place producing ydrogen gas and a solution of sodium hydroxide. These are not the ame substances as sodium and water.

Many medicine bottles contain an insoluble solid in water. The bottle has to be shaken before use to produce a suspension so that the solid is spread evenly throughout the bottle and the patient takes the correct amount of the solid.

Gardeners spray solutions containing copper sulphate onto some plants to protect them from disease.

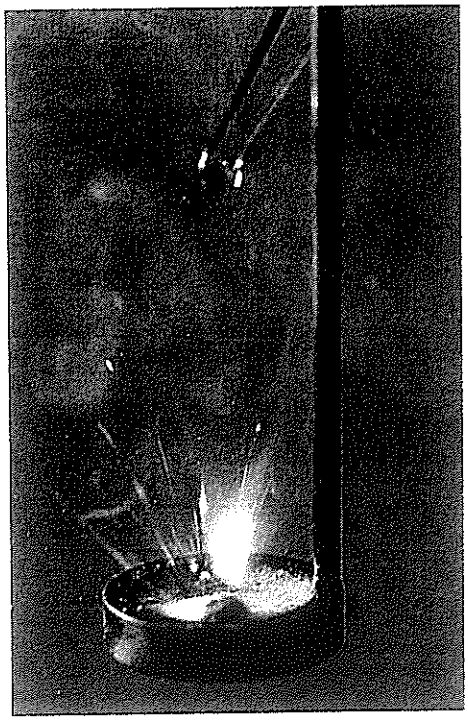


Figure 4.4 *Sodium reacting with water.*

2.4 Solutions

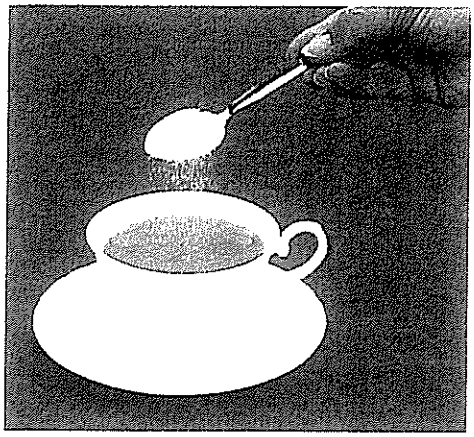


Figure 4.1 *Sugar dissolves in water.*

Have you ever wondered what happens to sugar when you add it to a cup of tea? Different liquids can dissolve different substances, but some substances will not dissolve in some liquids. Dissolving is a physical change.

What happens when something dissolves?

When sugar dissolves in a cup of tea, you can still taste the sugar. It is still there even though you cannot see it. When sugar dissolves, the moving water particles hit the sugar particles, knocking them into the water. This produces a mixture of water and sugar particles.

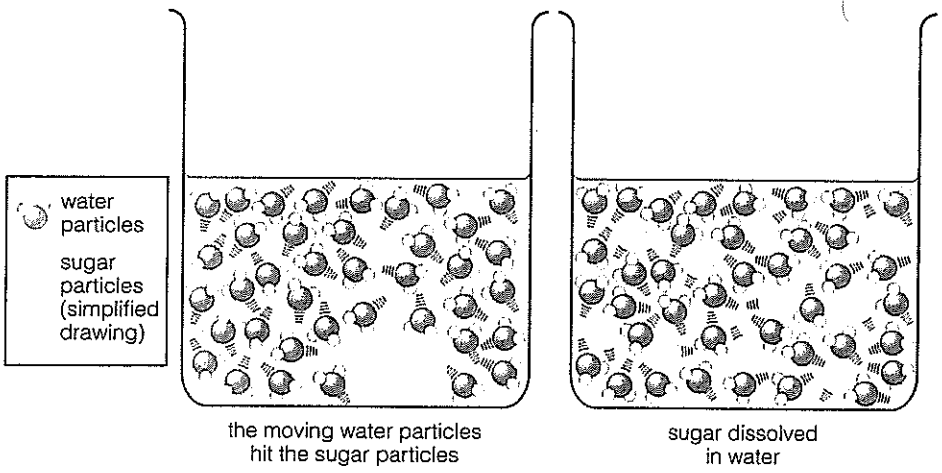


Figure 4.2 *Sugar particles dissolving in water.*

You can dissolve about 120 teaspoons of sugar in a tea cup of hot water (if you can keep it hot while you try!).

There are about 27 g of salt dissolved in one litre of sea water. It contains many other dissolved substances as well.

The sugar is said to be **soluble** in water because it dissolves in it. The sugar is known as the **solute** (the substance that dissolves) and the water as the **solvent** (the liquid it dissolves in). A **solution** is what you get when a solute dissolves in a solvent. Solutions are transparent (i.e. clear, you can see through them). There are many other solvents besides water, such as ethanol, propanone, turpentine and white spirit

Mass is conserved when a solution is formed. That is, the total mass of a solution is equal to the mass of the solvent plus the mass of the solute. For example, if 5 g of sugar dissolves in 100 g of water, the mass of the solution would be 105 g.

4 Diffusion happens faster in gases than in liquids. Why does this happen? Tick the correct explanation.

- ☐ The particles in gases are smaller than the particles in liquids.
- ☐ The particles in gases are bigger than the particles in liquids.
- ☐ The particles in gases are moving faster than the particles in liquids.
- ☐ The particles in gases are moving more slowly than the particles in liquids.

[observing]